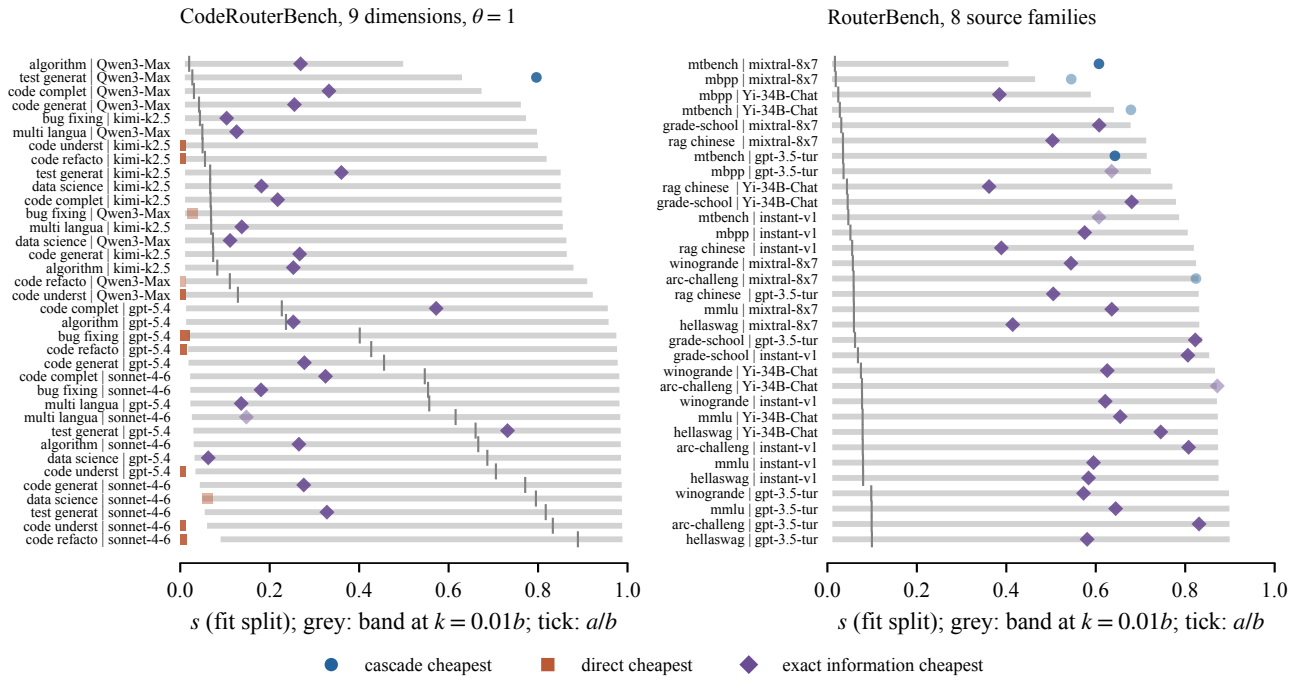
CodeRouterBench, 9 dimensions, $\theta = 1$
algorithm | Qwen3-Max
test generat | Qwen3-Max
code complet | Qwen3-Max
code generat | Qwen3-Max
bug fixing | kimi-k2.5
multi langua | Qwen3-Max
code underst | kimi-k2.5
code refacto | kimi-k2.5
test generat | kimi-k2.5
data science | kimi-k2.5
code complet | kimi-k2.5
bug fixing | Qwen3-Max
multi langua | kimi-k2.5
data science | Qwen3-Max
code generat | kimi-k2.5
algorithm | kimi-k2.5
code refacto | Qwen3-Max
code underst | Qwen3-Max
code complet | gpt-5.4
algorithm | gpt-5.4
bug fixing | gpt-5.4
code refacto | gpt-5.4
code generat | gpt-5.4
code complet | sonnet-4-6
bug fixing | sonnet-4-6
multi langua | gpt-5.4
multi langua | sonnet-4-6
test generat | gpt-5.4
algorithm | sonnet-4-6
data science | gpt-5.4
code underst | gpt-5.4
code generat | sonnet-4-6
data science | sonnet-4-6
test generat | sonnet-4-6
code underst | sonnet-4-6
code refacto | sonnet-4-6
0.0
0.2
0.4
0.6
0.8
1.0
s (fit split); grey: band at $k = 0.01b$; tick: a/b
RouterBench, 8 source families
mtbench | mixtral-8x7
mbpp | mixtral-8x7
mbpp | Yi-34B-Chat
mtbench | Yi-34B-Chat
grade-school | mixtral-8x7
rag chinese | mixtral-8x7
mtbench | gpt-3.5-tur
mbpp | gpt-3.5-tur
rag chinese | Yi-34B-Chat
grade-school | Yi-34B-Chat
mtbench | instant-v1
mbpp | instant-v1
rag chinese | instant-v1
winogrande | mixtral-8x7
arc-challeng | mixtral-8x7
rag chinese | gpt-3.5-tur
mmlu | mixtral-8x7
hellaswag | mixtral-8x7
grade-school | gpt-3.5-tur
grade-school | instant-v1
winogrande | Yi-34B-Chat
arc-challeng | Yi-34B-Chat
winogrande | instant-v1
mmlu | Yi-34B-Chat
hellaswag | Yi-34B-Chat
arc-challeng | instant-v1
mmlu | instant-v1
hellaswag | instant-v1
winogrande | gpt-3.5-tur
mmlu | gpt-3.5-tur
arc-challeng | gpt-3.5-tur
hellaswag | gpt-3.5-tur
0.0
0.2
0.4
0.6
0.8
1.0
s (fit split); grey: band at $k = 0.01b$; tick: a/b
cascade cheapest
direct cheapest
exact information cheapest